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NEET · BIOLOGY · CHAPTER-WISE PYQS

Sexual Reproduction in Flowering Plants

NEET Biology · Previous Year Questions with Answer Key

163	50	1988 – 2026	English
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MCQ (Single Correct Answer)	163 questions
Answer key	at the end

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MCQ (Single Correct Answer) 163 questions

Q1. RE-NEET 2026

Which of the following plants produces non-albuminous seeds?

(A) Pea (B) Wheat (C) Maize (D) Barley

Open solution

Q2. RE-NEET 2026

Which of the following in female gametophyte of an angiosperm helps in guiding the pollen tube for fertilizing the eggs?

(A) Polar nucleus (B) Antipodals (C) Synergids (D) Central cells

Open solution

Q3. RE-NEET 2026

If the diploid chromosome number of typical angiosperm is 36 , what would be the chromosome number in its endosperm?

(A) 72 (B) 18 (C) 36 (D) 54

Open solution

Q4. RE-NEET 2026

How many theca are present in each lobe of a typical bilobed angiosperm anther ?

(A) 12 (B) 2 (C) 6 (D) 8

Open solution

Q5. NEET 2026

Arrange the following in the correct developmental sequence related to microsporogenesis :

A. Microspore tetrads

B. Sporogenous tissue

C. Pollen grains

D. Pollen mother cells

Choose the correct answer from the options given below :

(A) D, A, C, B (B) B, D, A, C

(C) B, D, C, A (D) A, D, C, B

Open solution

Q6. NEET 2026

Which one of the following is a triploid cell?

(A) Synergid (B) Central cell

(C) Zygote (D) Primary endosperm cell

Open solution

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Q7. NEET 2026

Which one of the following types of pollination brings genetically different types of pollen grains to the stigma?

(A) Autogamy (B) Xenogamy (C) Geitonogamy (D) Cleistogamy

Open solution

Q8. NEET 2025

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : A typical unfertilised, angiosperm embryo sac at maturity is 8 nucleate and 7-celled.

Reason (R) : The egg apparatus has 2 polar nuclei.

In the light of the above statements, choose the correct answer from the options given below :

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Open solution

Q9. NEET 2025

Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Both wind and water pollinated flowers are not very colourful and do not produce nectar.

Reason (R) : The flowers produce enormous amount of pollen grains in wind and water pollinated flowers.

In the light of the above statements, choose the correct answer from the options given below:

(A) $\mathbf{A}$ is true but $\mathbf{R}$ is false

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Open solution

Q10. NEET 2025

How many meiotic and mitotic divisions need to occur for the development of a mature female gametophyte from the megaspore mother cell in an angiosperm plant?

(A) 1 Meiosis and 3 Mitosis (B) No Meiosis and 2 Mitosis

(C) 2 Meiosis and 3 Mitosis (D) 1 Meiosis and 2 Mitosis

Open solution

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Q11. NEET 2025

Open solution

Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Cells of the tapetum possess dense cytoplasm and generally have more than one nucleus.

Reason (R) : Presence of more than one nucleus in the tapetum increases the efficiency of nourishing the developing microspore mother cells.

In light of the above statements, choose the most appropriate answer from the options given below:

(A) A is true but $\mathbf{R}$ is false

(B) $\mathbf{A}$ is false but $\mathbf{R}$ is true

(C) Both $\mathbf{A}$ and $\mathbf{R}$ are true and $\mathbf{R}$ is the correct explanation of $\mathbf{A}$

(D) Both $\mathbf{A}$ and $\mathbf{R}$ are true but $\mathbf{R}$ is NOT the correct explanation of $\mathbf{A}$

Q12. NEET 2025

Open solution

Match List I with List II :

	List - I		List - II
A	Scutellum	I	Persistent nuclellus
B	Non-albuminous seed	II	Cotyledon of Monocot seed
C	Epiblast	III	Groundnut
D	Perisperm	IV	Rudimentary cotyledon

Choose the option with all correct matches.

(A) A-IV, B- III, C-I, D-II

(B) A-II, B- IV, C-III, D-I

(C) A-II, B- III, C-IV, D-I

(D) A-IV, B- III, C-II, D-I

Q13. NEET 2024 (Re-Examination)

Open solution

Which part of the ovule stores reserve food materials?

(A) Nucellus

(B) Integument

(C) Placenta

(D) Funicle

Q14. NEET 2024 (Re-Examination)

Open solution

Pollen grains remain preserved as fossils due to the presence of :

(A) Epidermal layer

(B) Tapetum

(C) Exine layer

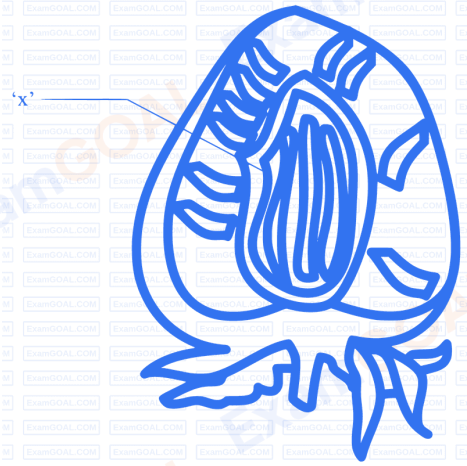
(D) Intine layer

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Q15. NEET 2024 (Re-Examination)

Open solution

The part marked as 'x' in the given figure is



(A) Endosperm (B) Thalamus (C) Endocarp (D) Mesocarp

Q16. NEET 2024

Open solution

Identify the set of correct statements:

A. The flowers of Vallisneria are colourful and produce nectar.

B. The flowers of water lily are not pollinated by water.

C. In most of water-pollinated species, the pollen grains are protected from wetting.

D. Pollen grains of some hydrophytes are long and ribbon like.

E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below.

(A) C, D and E only (B) A, B, C and D only

(C) A, C, D and E only (D) B, C, D and E only

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Q17. NEET 2024

Identify the correct description about the given figure :



Open solution

(A) Wind pollinated plant inflorescence showing flowers with well exposed stamens.

(B) Water pollinated flowers showing stamens with mucilaginous covering.

(C) Cleistogamous flowers showing autogamy.

(D) Compact inflorescence showing complete autogamy

Q18. NEET 2023 Manipur

In which of the following sets of families, the pollen grains are viable for months?

(A) Solanaceae, Poaceae and Liliaceae

(B) Brassicaceae, Liliaceae and Poaceae

(C) Rosaceae, Liliaceae and Poaceae

(D) Leguminosae, Solanaceae and Rosaceae

Open solution

Q19. NEET 2023 Manipur

Transfer of pollen grains from anther to stigma of another flower of same plant is known as :

(A) Geitonogamy

(B) Xenogamy

(C) Autogamy

(D) Cleistogamy

Open solution

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Q20. NEET 2023 Manipur

In angiosperms the correct sequence of events in formation of female gametophyte in the ovule is :

(A) 3 successive free nuclear divisions in functional megaspore.

(B) Degeneration of 3 megaspores.

(C) Meiotic division in megaspore mother cell.

(D) Migration of 3 nuclei towards each pole.

(E) Formation of wall resulting in seven celled embryosac.

Choose the correct answer from the options given below :

(A) (A), (B), (C), (D), (E)
(B) (C), (E), (A), (D), (B)
(C) (B), (C), (A), (D), (E)
(D) (C), (B), (A), (D), (E)

Open solution

Q21. NEET 2023

What is the function of tassels in the corn cob?

(A) To trap pollen grains
(B) To disperse pollen grains
(C) To protect seeds
(D) To attract insects

Open solution

Q22. NEET 2023

Large, colourful, fragrant flowers with nectar are seen in

(A) Bird pollinated plants
(B) Bat pollinated plants
(C) Wind pollinated plants
(D) Insect pollinated plants

Open solution

Q23. NEET 2023

In angiosperm, the haploid, diploid and triploid structures of a fertilized embryo sac sequentially are :

(A) Antipodals, synergids, and primary endosperm nucleus
(B) Synergids, Zygote and Primary endosperm nucleus
(C) Synergids, antipodals and Polar nuclei
(D) Synergids, Primary endosperm nucleus and zygote

Open solution

Q24. NEET 2022 Phase 2

The residual persistent part which forms the perisperm in the seeds of beet is

(A) Integument **(B)** Calyx **(C)** Endosperm **(D)** Nucellus

Open solution

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Q25. NEET 2022 Phase 2

To ensure that only the desired pollens fall on the stigma in artificial hybridization process

(a) the female flower buds of plant producing unisexual flowers need not be bagged.

(b) there is no need to emasculate unisexual flowers of selected female parent

(c) emasculated flowers are to be bagged immediately after cross pollination

(d) emasculated flowers are to be bagged after removal of anthers

(e) bisexual flowers, showing protogyny are never selected for cross

Choose the correct answer from the options given belows

(A) (a), (d) and (e) only

(B) (a), (b) and (c) only

(C) (b), (c) and (d) only

(D) (b), (c) and (e) only

Open solution

Q26. NEET 2022 Phase 2

In general the egg apparatus of embryo sac in angiosperm consists of

(A) One egg cell, two synergids, two antipodal cells, two Polar nuclei

(B) One egg cell, two synergids, three antipodal cells, two Polar nuclei

(C) One egg cell, two synergids, two antipodal cells, three Polar nuclei

(D) One egg cell, three synergids, two antipodal cells, two Polar nuclei

Open solution

Q27. NEET 2022 Phase 1

Given below are two statements:

Statement I : Cleistogamous flowers are invariably autogamous

Statement II : Cleistogamy is disadvantageous as there is no chance for cross pollination

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Open solution

Q28. NEET 2022 Phase 1

Identify the incorrect statement related to Pollination:

(A) Pollination by water is quite rare in flowering plants

(B) Pollination by wind is more common amongst abiotic pollination

(C) Flowers produce foul odours to attract flies and beetles to get pollinated

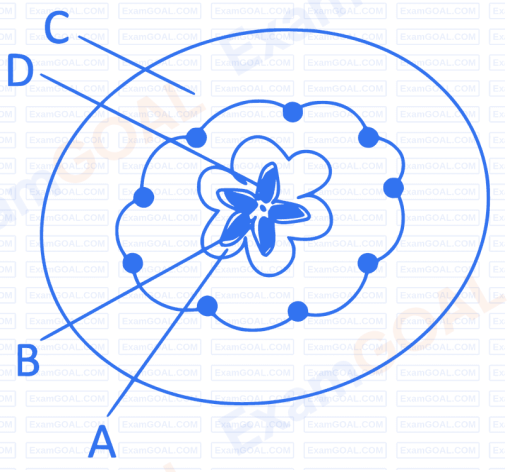
(D) Moths and butterflies are the most dominant pollinating agents among insects

Open solution

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Q29. NEET 2022 Phase 1

Which part of the fruit, labelled in the given figure makes it a false fruit?



Open solution

(A) A → Mesocarp (B) B → Endocarp (C) C → Thalamus (D) D → Seed

Q30. NEET 2021

The term used for transfer of pollen grains from anthers of one plant to stigma of a different plant which, during pollination, brings genetically different types of pollen grains to stigma, is :

(A) Cleistogamy (B) Xenogamy (C) Geitonogamy (D) Chasmogamy

Open solution

Q31. NEET 2021

A typical angiosperm embryo sac at maturity is :

(A) 8-nucleate and 8-celled
(B) 8-nucleate and 7-celled
(C) 7-nucleate and 8-celled
(D) 7-nucleate and 7-celled

Open solution

Q32. NEET 2021

In some members of which of the following pairs of families, pollen grains retain their viability for months after release?

(A) Rosaceae : Leguminosae
(B) Poaceae : Rosaceae
(C) Poaceae : Leguminosae
(D) Poaceae : Solanaceae

Open solution

Q33. NEET 2020 Phase 2

In some plants thalamus contributes to fruit formation. Such fruits are termed as :

(A) Parthenocarpic fruit
(B) False fruits
(C) Aggregate fruits
(D) True fruits

Open solution

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Q34. NEET 2020 Phase 2

Which of the following is **incorrect** for wind-pollinated plants ?

(A) Pollen grains are light and non-sticky

(B) Well exposed stamens and stigma

(C) Many ovules in each ovary

(D) Flowers are small and not brightly coloured

Open solution

Q35. NEET 2020 Phase 1

The body of the ovule is fused within the funicle at

(A) Chalaza (B) Hilum (C) Micropyle (D) Nucellus

Open solution

Q36. NEET 2020 Phase 1

The plant parts which consist of two generations - one within the other

(i) Pollen grains inside the anther

(ii) Germinated pollen grain with two male gametes

(iii) Seed inside the fruit

(iv) Embryo sac inside the ovule

(A) (i) and (iv) (B) (i) only

(C) (i), (ii) and (iii) (D) (iii) and (iv)

Open solution

Q37. NEET 2019 (Odisha)

Which is the most common type of embryo sac in angiosperms?

(A) Bisporic with two sequential mitotic divisions

(B) Tetrasporic with one mitotic stage of divisions

(C) Monosporic with three sequential mitotic divisions

(D) Monosporic with two sequential mitotic divisions

Open solution

Q38. NEET 2019 (Odisha)

What type of pollination takes place in *Vallisneria*?

(A) Male flowers are carried by water currents to female flowers at surface of water.

(B) Pollination occurs in submerged condition by water.

(C) Flowers emerge above surface of water and pollination occurs by insects.

(D) Flowers emerge above water surface and pollen is carried by wind.

Open solution

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Q39. NEET 2019 (Odisha)

In which one of the following, both autogamy and geitonogamy are prevented?

(A) Maize (B) Wheat (C) Papaya (D) Castor

Open solution

Q40. NEET 2019

In some plants, the female gamete develops into embryo without fertilization. This phenomenon is known as :

(A) Syngamy (B) Parthenocarpy (C) Parthenogenesis (D) Autogamy

Open solution

Q41. NEET 2019

Which one of the following statements regarding post-fertilization development in flowering plants is incorrect?

(A) Central cell develops into endosperm (B) Ovules develop into embryo sac

(C) Ovary develops into fruit (D) Zygote develops into embryo

Open solution

Q42. NEET 2019

Persistent nucellus in the seed is known as

(A) Hilum (B) Tegmen (C) Chalaza (D) Perisperm

Open solution

Q43. NEET 2019

What is the fate of the male gametes discharged in the synergid?

(A) One fuses with the egg, other (s) fuse (s) with synergid nucleus.

(B) One fuses with the egg and other fuses with central cell nuclei.

(C) One fuses with the egg, other (s) degenerate (s) in the synergid.

(D) All fuse with the egg.

Open solution

Q44. NEET 2018

Which of the following has proved helpful in preserving pollen as fossils?

(A) Pollenkitt (B) Cellulosic intine

(C) Oil content (D) Sporopollenin

Open solution

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Q45. NEET 2018

Double fertilization is

(A) fusion of two male gametes of a pollen tube with two different eggs

(B) fusion of one male gamete with two polar nuclei

(C) fusion of two male gametes with one egg

(D) syngamy and triple fusion

Open solution

Q46. NEET 2018

Pollen grains can be stored for several years in liquid nitrogen having a temperature of

(A) -120°C (B) -80°C (C) -196°C (D) -160°C

Open solution

Q47. NEET 2017

The morphological nature of the edible part of coconut is

(A) cotyledon (B) endosperm (C) pericarp (D) perisperm.

Open solution

Q48. NEET 2017

Functional megaspore in an angiosperm develops into an

(A) endosperm (B) embryo sac (C) embryo (D) ovule.

Open solution

Q49. NEET 2017

Attractants and rewards are required for

(A) entomophily (B) hydrophily (C) cleistogamy (D) anemophily.

Open solution

Q50. NEET 2017

Flowers which have single ovule in the ovary and are packed into inflorescence are usually pollinated by

(A) bee (B) wind (C) bat (D) water

Open solution

Q51. NEET 2017

A dioecious flowering plant prevents both

(A) autogamy and geitonogamy (B) geitonogamy and xenogamy

(C) cleistogamy and xenogamy (D) autogamy and xenogamy.

Open solution

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Q52. NEET 2017

Double fertilisation is exhibited by

(A) algae (B) fungi (C) angiosperms (D) gymnosperms.

Open solution

Q53. NEET 2016 Phase 2

In majority of angiosperms

(A) egg has a filiform apparatus

(B) there are numerous antipodal cells

(C) reduction division occurs in the megaspore mother cells

(D) a small central cell is present in that embryo sac.

Open solution

Q54. NEET 2016 Phase 2

Pollination in water hyacinth and water lily is brought about by the agency of

(A) water (B) insects or wind (C) birds (D) bats.

Open solution

Q55. NEET 2016 Phase 2

The ovule of an angiosperm is technically equivalent to

(A) megasporangium (B) megasporophyll

(C) megaspore mother cell (D) megaspore.

Open solution

Q56. NEET 2016 Phase 1

The coconut water from tender coconut represents

(A) free nuclear proembryo (B) free nuclear endosperm

(C) endocarp (D) fleshy mesocarp.

Open solution

Q57. NEET 2016 Phase 1

Which one of the following statements is not true?

(A) Pollen grains of many species cause severe allergies.

(B) Stored pollen in liquid nitrogen can be used in the crop breeding programmes.

(C) Tapetum helps in the dehiscence of anther.

(D) Exine of pollen grains is made up of sporopollenin.

Open solution

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Q58. NEET 2016 Phase 1

Seed formation without fertilisation in flowering plants involves the process of

(A) somatic hybridisation
(B) apomixis
(C) sporulation
(D) budding.

Open solution

Q59. NEET 2016 Phase 1

Proximal end of the filament of stamen is attached to the

(A) Placenta
(B) thalamus or petal
(C) anther
(D) connective.

Open solution

Q60. NEET 2016 Phase 1

Which of the following statements is not correct?

(A) Pollen germination and pollen tube growth are regulated by chemical components of pollen interacting with those of the pistil.
(B) Some reptiles have also been reported as pollinators in some plant species.
(C) Pollen grains of many species can germinate on the stigma of a flower, but only one pollen tube of the same species grows into the style.
(D) Insects that consume pollen or nectar without bringing about pollination are called pollen/ nectar robbers.

Open solution

Q61. AIPMT 2015

Filiform apparatus is characteristic feature of

(A) aleurone cell
(B) synergids
(C) generative cell
(D) nucellar embryo.

Open solution

Q62. AIPMT 2015

In angiosperms, microsporogenesis and mega-sporogenesis

(A) involve meiosis
(B) occur in ovule
(C) occur in anther
(D) form gametes without further divisions.

Open solution

Q63. AIPMT 2015

Flowers are unisexual in

(A) China rose
(B) onion
(C) pea
(D) cucumber.

Open solution

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Q64. AIPMT 2015

Coconut water from a tender coconut is

(A) innermost layers of the seed coat
(B) degenerated nucellus
(C) immature embryo
(D) free nuclear endosperm.

Open solution

Q65. AIPMT 2015

Which one of the following fruits is parthenocarpic?

(A) Jackfruit (B) Banana (C) Brinjal (D) Apple

Open solution

Q66. AIPMT 2015

Male gametophyte in angiosperms produces

(A) single sperm and two vegetative cells
(B) three sperms
(C) two sperms and a vegetative cell
(D) single sperm and a vegetative cell.

Open solution

Q67. AIPMT 2015 Cancelled Paper

Which of the following are the important floral rewards to the animal pollinators ?

(A) Floral fragrance and calcium crystals
(B) Protein pellicle and stigmatic exudates
(C) Colour and large size of flower
(D) Nectar and pollen grains

Open solution

Q68. AIPMT 2015 Cancelled Paper

Which one of the following may require pollinators, but is genetically similar to autogamy?

(A) Apogamy (B) Cleistogamy (C) Geitonogamy (D) Xenogamy

Open solution

Q69. AIPMT 2015 Cancelled Paper

Which one of the following statements is not true?

(A) The flowers pollinated by flies and bats secrete foul odour to attract them.
(B) Honey is made by bees by digesting pollen collected from flowers.
(C) Pollen grains are rich in nutrients and they are used in the form of tablets and syrups.
(D) Pollen grains of some plants cause severe allergies and bronchial afflictions in some people.

Open solution

Q70. AIPMT 2015 Cancelled Paper

Transmission tissue is characteristic feature of

(A) dry stigma (B) wet stigma (C) hollow style (D) solid style.

Open solution

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Q71. AIPMT 2015 Cancelled Paper

The hilum is a scar on the

(A) seed, where funicle was attached
(B) fruit, where style was present
(C) fruit, where it was attached to pedicel
(D) seed, where micropyle was present

Open solution

Q72. AIPMT 2014

Geitonogamy involves

(A) fertilization of a flower by the pollen from another flower of the same plant
(B) fertilization of a flower by the pollen from the same flower
(C) fertilization of a flower by the pollen from a flower of another plant in the same population
(D) fertilization of a flower by the pollen from a flower of another plant belonging to a distant population.

Open solution

Q73. AIPMT 2014

Pollen tablets are available in the market for

(A) in *vitro* fertilization
(B) breeding programmes
(C) supplementing food
(D) *ex situ* conservation.

Open solution

Q74. AIPMT 2014

Function of filiform apparatus is to

(A) recognize the suitable pollen at stigma
(B) stimulate division of generative cell
(C) produce nectar
(D) guide the entry of pollen tube.

Open solution

Q75. AIPMT 2014

Non-albuminous seed is produced in

(A) maize
(B) castor
(C) wheat
(D) pea.

Open solution

Q76. NEET 2013 (Karnataka)

Animal vectors are required for pollination in

(A) *Vallisneria*
(B) mulberry
(C) cucumber
(D) maize.

Open solution

Q77. NEET 2013 (Karnataka)

Albuminous seeds store their reserve food mainly in

(A) endosperm
(B) cotyledons
(C) hypocotyl
(D) perisperm.

Open solution

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Q78. NEET 2013 (Karnataka)

Megaspores are produced from the megaspore mother cells after

(A) mitotic division
(B) formation of thick wall
(C) differentiation
(D) meiotic division.

Open solution

Q79. NEET 2013 (Karnataka)

Which one of the following statements is correct?

(A) Cleistogamous flowers are always autogamous.
(B) Xenogamy occurs only by wind pollination.
(C) Chasmogamous flowers do not open at all.
(D) Geiotonogamy involves the pollen and stigma of flowers of different plants.

Open solution

Q80. NEET 2013 (Karnataka)

Which of the following statements is correct?

(A) Sporopollenin can be degraded by enzymes.
(B) Sporopollenin is made up of inorganic materials.
(C) Sporopollenin can withstand high temperatures as well as strong acids and alkalis.
(D) Sporopollenin can withstand high temperatures but not strong acids.

Open solution

Q81. NEET 2013

Meiosis takes place in

(A) gemmule
(B) megaspore
(C) meiocyte
(D) conidia.

Open solution

Q82. NEET 2013

Seed coat is not thin, membranous in

(A) groundnut
(B) gram
(C) maize
(D) coconut.

Open solution

Q83. NEET 2013

Perisperm differs from endosperm in

(A) being a diploid tissue
(B) its formation by fusion of secondary nucleus with several sperms
(C) being a haploid tissue
(D) having no reserve food.

Open solution

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Q84. NEET 2013

Advantage of cleistogamy is

(A) no dependence on pollinators
(B) vivipary
(C) higher genetic variability
(D) more vigorous offspring.

Open solution

Q85. NEET 2013

Megasporangium is equivalent to

(A) nucellus
(B) ovule
(C) embryo sac
(D) fruit.

Open solution

Q86. NEET 2013

Which one of the following statements is correct?

(A) Endothecium produces the microspores.
(B) Tapetum nourishes the developing pollen.
(C) Hard outer layer of pollen is called intine.
(D) Sporogenous tissue is haploid.

Open solution

Q87. NEET 2013

Product of sexual reproduction generally generates

(A) new genetic combination leading to variation
(B) large biomass
(C) longer viability of seeds
(D) prolonged dormancy.

Open solution

Q88. AIPMT 2012 Mains

What is the function of germ pore?

(A) Emergence of radicle
(B) Absorption of water for seed germination
(C) Initiation of pollen tube
(D) Release of male gametes

Open solution

Q89. AIPMT 2012 Mains

Which one of the following statements is wrong?

(A) When pollen is shed at two-celled stage, double fertilization does not take place.
(B) Vegetative cell is larger than generative cell.
(C) Pollen grains in some plants remain viable for months.
(D) Intine is made up of cellulose and pectin.

Open solution

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Q90. AIPMT 2012 Mains

Plants with ovaries having only one or a few ovules, are generally pollinated by

(A) bees (B) butterflies (C) birds (D) wind.

Open solution

Q91. AIPMT 2012 Prelims

Both, autogamy and geitonogamy are prevented in

(A) Papaya (B) cucumber (C) castor (D) maize.

Open solution

Q92. AIPMT 2012 Prelims

An organic substance that can withstand environmental extremes and cannot be degraded by any enzyme is

(A) cuticle (B) sporopollenin (C) lignin (D) cellulose.

Open solution

Q93. AIPMT 2012 Prelims

Which one of the following is correctly matched?

(A) Onion—Bulb (B) Ginger—Sucker

(C) Chlamydomonas—Conidia (D) Yeast—Zoospores

Open solution

Q94. AIPMT 2012 Prelims

Even in absence of pollinating agents seed-setting is assured in

(A) *Commelina* (B) *Zostera* (C) *Salvia* (D) fig.

Open solution

Q95. AIPMT 2011 Mains

In angiosperms, functional megaspore develops into

(A) embryo sac (B) ovule (C) endosperm (D) pollen sac.

Open solution

Q96. AIPMT 2011 Mains

What is common between vegetative reproduction and apomixis?

(A) Both are applicable to only dicot plants

(B) Both bypass the flowering phase

(C) Both occur round the year

(D) Both produce progeny identical to the parent.

Open solution

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Q97. AIPMT 2011 Prelims

Filiform apparatus is a characteristic feature of

(A) suspensor (B) egg (C) synergid (D) zygote.

Open solution

Q98. AIPMT 2011 Prelims

Nucellar polyembryony is reported in species of

(A) *Citrus* (B) *Gossypium* (C) *Triticum* (D) Brassica.

Open solution

Q99. AIPMT 2011 Prelims

Which one of the following pollinations is autogamous?

(A) Geitonogamy (B) Xenogamy (C) Chasmogamy (D) Cleistogamy

Open solution

Q100. AIPMT 2011 Prelims

Wind pollination is common in

(A) legumes (B) lilies (C) grasses (D) orchids.

Open solution

Q101. AIPMT 2010 Mains

Vegetative propagation in *Pistia* occurs by

(A) stolon (B) offset (C) runner (D) sucker.

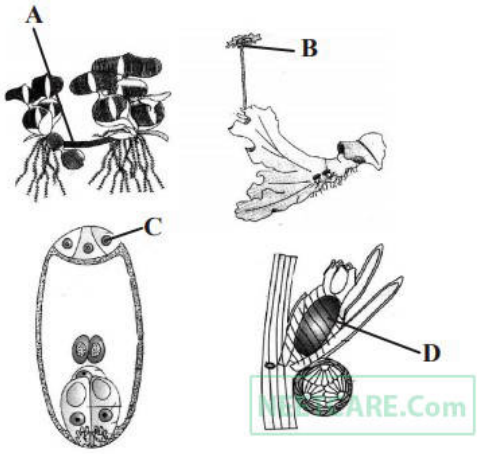
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Q102. AIPMT 2010 Mains

Examine the figures (A-D) given below and select the right option out of (a – d), in which all the four structures A, B, C and D are identified correctly.

Open solution



(A) **A** → offset, **B** → Antheridiophore, **C** → Antipodals, **D** → Oogonium

(B) **A** → Rhizome, **B** → Sporangiophore, **C** → Polar cell, **D** → Globule

(C) **A** → Sucker, **B** → Seta, **C** → Megaspore mother cell , **D** → Gemma cup

(D) **A** → Runner, **B** → Archegoniophore, **C** → Synergid, **D** → Antheridium

Q103. AIPMT 2010 Prelims

Apomictic embryos in *Citrus* arise form

(A) synergids

(B) maternal sporophytic tissue in ovule

(C) antipodal cells

(D) diploid egg.

Open solution

Q104. AIPMT 2010 Prelims

Transfer of pollen grains from the anther to the stigma of another flower of the same plant is called

(A) xenogamy (B) geitonogamy (C) karyogamy (D) autogamy.

Open solution

Q105. AIPMT 2010 Prelims

Wind pollinated flowers are

(A) small, brightly coloured, producing large number of pollen grains

(B) small, producing large number of dry pollen grains

(C) large producing abundant nectar and pollen

(D) small, producing nectar and dry pollen.

Open solution

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Q106. AIPMT 2009

Vegetative propagation in mint occurs by

(A) offset (B) rhizome (C) sucker (D) runner.

Open solution

Q107. AIPMT 2008

Which one of the following pairs of plant structures has haploid number of chromosomes ?

(A) Nucellus and antipodal cells

(B) Egg nucleus and secondary nucleus

(C) Megaspore mother cell and antipodal cells

(D) Egg cell and antipodal cells

Open solution

Q108. AIPMT 2008

What does the filiform apparatus do at the entrance into ovule?

(A) It brings about opening of the pollen tube.

(B) It guides pollen tube from a synergid to egg.

(C) It helps in the entry of pollen tube into a synergid.

(D) It prevents entry of more than one pollen tube into the embryo sac.

Open solution

Q109. AIPMT 2008

Unisexuality of flowers prevents

(A) geitonogamy, but not xenogamy (B) autogamy and geitonogamy

(C) autogamy, but not geitonogamy (D) both geitonogamy and xenogamy.

Open solution

Q110. AIPMT 2008

Which one of the following is resistant to enzyme action?

(A) Pollen exine (B) Leaf cuticle (C) Cork (D) Wood fibre

Open solution

Q111. AIPMT 2007

Male gametes in angiosperms are formed by the division of

(A) generative cell (B) vegetative cell

(C) microspore mother cell (D) microspore.

Open solution

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Q112. AIPMT 2007

Two plants can be conclusively said to belong to the same species if they

(A) have more than 90 percent similar genes

(B) look similar and possess identical secondary metabolites

(C) have same number of chromosomes

(D) can reproduce freely with each other and form seeds.

Open solution

Q113. AIPMT 2007

Which one of the following is surrounded by a callose wall?

(A) Male gamete (B) Egg

(C) Pollen grain (D) Microspore mother cell

Open solution

Q114. AIPMT 2006

In a cereal grain the single cotyledon of embryo is represented by

(A) coleoptile (B) coleorhiza (C) scutellum (D) prophyll.

Open solution

Q115. AIPMT 2006

The arrangement of the nuclei in a normal embryo sac in the dicot plants is

(A) 3 + 3 +2 (B) 2 + 4 + 2 (C) 3 + 2 + 3 (D) 2 + 3 + 3

Open solution

Q116. AIPMT 2006

What would be the number of chromosomes in the cells of the aleurone layer in a plant species with 8 chromosomes in its synergids?

(A) 16 (B) 24 (C) 32 (D) 8

Open solution

Q117. AIPMT 2005

In a type of apomixis known as adventive embryony, embryos develop directly from the

(A) nucellus or integuments (B) zygote

(C) synergids or antipodals in an embryo sac (D) accessory embryo sacs in the ovule.

Open solution

Q118. AIPMT 2005

Which one of the following represents an ovule, where the embryo sac becomes horse-shoe shaped and the funiculus and micropyle are close to each other?

(A) Amphitropous (B) Circinotropous (C) Atropous (D) Anatropous

Open solution

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Q119. AIPMT 2005

Through which cell of the embryo sac, does the pollen tube enter the embryo sac?

(A) Egg cell
(B) Persistent synergid
(C) Degenerated synergids
(D) Central cell

Open solution

Q120. AIPMT 2004

When a diploid female plant is crossed with a tetraploid male, the ploidy of endosperm cells in the resulting seed is

(A) tetraploidy
(B) pentaploidy
(C) diploidy
(D) triploidy

Open solution

Q121. AIPMT 2004

An ovule which becomes curved so that the nucellus and embryo sac lie at right angles to the funicle is

(A) hemitropous
(B) campylotropous
(C) anatropous
(D) orthotropous.

Open solution

Q122. AIPMT 2004

Anthesis is a phenomenon which refers to

(A) reception of pollen by stigma
(B) formation of pollen
(C) development of anther
(D) opening of flower bud.

Open solution

Q123. AIPMT 2003

In a flowering plant, archesporium gives rise to

(A) only the wall of the sporangium
(B) both wall and the sporogenous cells
(C) wall and the tapetum
(D) only tapetum and sporogenous cells.

Open solution

Q124. AIPMT 2002

In angiosperms pollen tube liberate their male gametes into the

(A) central cell
(B) antipodal cells
(C) egg cell
(D) synergids.

Open solution

Q125. AIPMT 2002

In angiosperm all the four microspores of tetrad are covered by a layer which is formed by

(A) pectocellulose
(B) callose
(C) cellulose
(D) sporopollenin.

Open solution

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Q126. AIPMT 2002

What is the direction of micropyle in anatropous ovule?

(A) Upwards (B) Downward (C) Right (D) Left

Open solution

Q127. AIPMT 2001

Adventive embryony in *Citrus* is due to

(A) nucellus (B) integuments (C) zygotic embryo (D) fertilized egg.

Open solution

Q128. AIPMT 2001

In grasses what happens in microspore mother cell for the formation of mature pollen grains?

(A) One meiotic and two mitotic divisions (B) One meiotic and one mitotic divisions
(C) One meiotic (D) One Mitotic division

Open solution

Q129. AIPMT 2001

Anemophily type of pollination is found in

(A) *Salvia* (B) bottle brush (C) *Vallisneria* (D) coconut.

Open solution

Q130. AIPMT 2000

Endosperm is formed during the double fertilization by

(A) two polar nuclei and one male gamete (B) one polar nuclei and one male gamete
(C) ovum and male gamete (D) two polar nuclei and two male gametes.

Open solution

Q131. AIPMT 2000

Eight nucleated embryo sac is

(A) only monosporic (B) only bisporic (C) only tetrasporic (D) any of these.

Open solution

Q132. AIPMT 1998

An interesting modification of flower shape for insect pollination occurs in some orchids in which a male insect mistakes the pattern on the orchid flower for the female species and tries to copulate with it, thereby pollinating the flower. This phenomenon is called

(A) mimicry (B) pseudocopulation
(C) pseudopollination (D) pseudoparthenocarpy

Open solution

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Q133. AIPMT 1998

The role of double fertilization in angiosperms is to produce

(A) endosperm (B) integuments (C) cotyledons (D) endocarp

Open solution

Q134. AIPMT 1997

If an angiospermic male plant is diploid and female plant tetraploid, the ploidy level of endosperm will be

(A) tetraploid (B) pentaploid (C) haploid (D) triploid

Open solution

Q135. AIPMT 1996

The role of double fertilization in angiosperms is to produce

(A) cotyledons (B) endocarp (C) endosperm (D) integuments

Open solution

Q136. AIPMT 1996

If there are 4 cells in anthers, what will be the number of pollen grains?

(A) 16 (B) 12 (C) 8 (D) 4

Open solution

Q137. AIPMT 1995

The anthesis is a phenomenon, which refers to

(A) development of anthers (B) opening of flower bud
(C) stigma receptors (D) all of these

Open solution

Q138. AIPMT 1995

In an angiosperm, how many microspore mother cells are required to produce 100 pollen grains?

(A) 75 (B) 100 (C) 25 (D) 50

Open solution

Q139. AIPMT 1995

The polyembryony commonly occurs in

(A) tomato (B) potato (C) *Citrus* (D) turmeric

Open solution

Q140. AIPMT 1994

When pollen of a flower is transferred to the stigma of another flower of the same plant, the pollination is referred to as

(A) autogamy (B) geitonogamy (C) xenogamy (D) allogamy

Open solution

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Q141. AIPMT 1994

Embryo sac represents

(A) megaspore
(B) megagametophyte
(C) megasporophyll
(D) megagamete

Open solution

Q142. AIPMT 1993

Number of meiotic divisions required to produce 200/400 seeds of pea would be

(A) 200/400
(B) 400/800
(C) 300/600
(D) 250/500

Open solution

Q143. AIPMT 1993

Double fertilization is characteristic of

(A) angiosperms
(B) anatropous
(C) gymnosperms
(D) bryophytes

Open solution

Q144. AIPMT 1993

Ovule is straight with funiculus, embryo sac, chalaza and micropyle lying on one straight line. It is

(A) orthotropous
(B) anatropous
(C) campylotropous
(D) amphitropous

Open solution

Q145. AIPMT 1993

Study of formation, growth and development of new individual from an egg is

(A) apomixis
(B) embryology
(C) embryogeny
(D) cytology

Open solution

Q146. AIPMT 1992

Meiosis is best observed in dividing

(A) cells of apical meristem
(B) cells of lateral meristem
(C) microspores and anther wall
(D) microsporocytes

Open solution

Q147. AIPMT 1991

Double fertilization is fusion of

(A) two eggs
(B) two eggs and polar nuclei with pollen nuclei
(C) one male gamete with egg and other with synergid
(D) one male gamete with egg and other with secondary nucleus

Open solution

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Q148. AIPMT 1991

Point out the odd one.

(A) Nucellus (B) Embryo sac (C) Micropyle (D) Pollen grain

Open solution

Q149. AIPMT 1991

Which of the following pair have haploid structures?

(A) Nucellus and antipodal cells

(B) Antipodal cells and egg cell

(C) Antipodal cells and megaspore mother cell

(D) Nucellus and primary endosperm nucleus

Open solution

Q150. AIPMT 1991

Embryo sac occurs in

(A) embryo (B) axis part of embryo

(C) ovule (D) endosperm

Open solution

Q151. AIPMT 1991

Pollination occurs in

(A) bryophytes and angiosperms (B) pteridophytes and angiosperms

(C) angiosperms and gymnosperms (D) angiosperms and fungi

Open solution

Q152. AIPMT 1990

Entry of pollen tube through micropyle is

(A) chalazogamy (B) mesogamy (C) porogamy (D) pseudogamy

Open solution

Q153. AIPMT 1990

Sperm and egg nuclei fuse due to

(A) base pairing of their DNA and RNA

(B) formation of hydrogen bonds

(C) mutual attraction due to differences in electrical charges

(D) attraction of their protoplasts

Open solution

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Q154. AIPMT 1990

Female gametophyte of angiosperms is represented by

(A) ovule
(B) megaspore mother cell
(C) embryo sac
(D) nucellus

Open solution

Q155. AIPMT 1990

Male gametophyte of angiosperms/monocots is

(A) microsporangium (B) nucellus (C) microspore (D) stamen

Open solution

Q156. AIPMT 1989

Generative cell was destroyed by laser but a normal pollen tube was still formed because

(A) vegetative cell is not damaged
(B) contents of killed generative cell stimulate pollen growth
(C) laser beam stimulates growth of pollen tube
(D) the region of emergence of pollen tube is not harmed

Open solution

Q157. AIPMT 1989

Nucellar embryo is

(A) amphimictic haploid
(B) amphimictic diploid
(C) apomictic haploid
(D) apomictic diploid

Open solution

Q158. AIPMT 1989

Perisperm is

(A) remnant of endosperm
(B) persistent nucellus
(C) peripheral part of endosperm
(D) disintegrated secondary nucleus

Open solution

Q159. AIPMT 1988

Double fertilization and triple fusion were discovered by

(A) Hofmeister
(B) Nawaschin and Guignard
(C) Leeuwenhoek
(D) Strasburger

Open solution

Q160. AIPMT 1988

Perisperm is

(A) remnant of endosperm
(B) persistent nucellus
(C) peripheral part of endosperm
(D) disintegrated secondary nucleus

Open solution

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Q161. AIPMT 1988

Total number of meiotic divisions required for forming 100 zygotes/100 grains of wheat is

(A) 100 (B) 75 (C) 125 (D) 50

Open solution

Q162. AIPMT 1988

Male gametophyte of angiosperms is shed at

(A) four celled pollen grain (B) three celled pollen grain

(C) microspore mother cell (D) anther

Open solution

Q163. AIPMT 1988

Formation of gametophyte directly from sporophyte without meiosis is

(A) apospory (B) apogamy (C) parthenogenesis (D) amphimixis

Open solution

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Answer Key

1. **A**	2. **C**	3. **D**	4. **B**	5. **B**
6. **D**	7. **B**	8. **A**	9. **D**	10. **A**
11. **C**	12. **C**	13. **A**	14. **C**	15. **B**
16. **D**	17. **A**	18. **D**	19. **A**	20. **D**
21. **B**	22. **D**	23. **B**	24. **D**	25. **C**
26. **B**	27. **A**	28. **D**	29. **C**	30. **B**
31. **B**	32. **A**	33. **B**	34. **C**	35. **B**
36. **A**	37. **C**	38. **A**	39. **C**	40. **C**
41. **B**	42. **D**	43. **B**	44. **D**	45. **D**
46. **C**	47. **B**	48. **B**	49. **A**	50. **B**
51. **A**	52. **C**	53. **C**	54. **B**	55. **A**
56. **B**	57. **C**	58. **B**	59. **B**	60. **C**
61. **B**	62. **A**	63. **D**	64. **D**	65. **B**
66. **C**	67. **D**	68. **C**	69. **B**	70. **D**
71. **A**	72. **A**	73. **C**	74. **D**	75. **D**
76. **C**	77. **A**	78. **D**	79. **A**	80. **C**
81. **C**	82. **D**	83. **A**	84. **A**	85. **B**
86. **B**	87. **A**	88. **C**	89. **A**	90. **D**
91. **A**	92. **B**	93. **A**	94. **A**	95. **A**
96. **D**	97. **C**	98. **A**	99. **D**	100. **C**
101. **B**	102. **A**	103. **B**	104. **B**	105. **B**
106. **C**	107. **D**	108. **C**	109. **C**	110. **A**
111. **A**	112. **D**	113. **D**	114. **C**	115. **C**
116. **B**	117. **A**	118. **A**	119. **C**	120. **A**
121. **A**	122. **D**	123. **B**	124. **D**	125. **B**
126. **B**	127. **A**	128. **A**	129. **D**	130. **A**
131. **D**	132. **B**	133. **A**	134. **B**	135. **C**
136. **A**	137. **B**	138. **C**	139. **C**	140. **B**
141. **B**	142. **D**	143. **A**	144. **A**	145. **B**
146. **D**	147. **D**	148. **D**	149. **B**	150. **C**
151. **C**	152. **C**	153. **D**	154. **C**	155. **C**

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156. **A**	157. **D**	158. **B**	159. **B**	160. **B**
161. **C**	162. **B**	163. **A**		

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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